Getting Started with Power BI

30-Minute Hands-On Exercise | Week 9 - Section 1A

Learning Objectives: Identify Power BI ecosystem components | Explain the role of xVelocity and columnar storage | Distinguish key Power BI tools | Describe how DAX fits into the data model

Instructions: Complete all four parts in 30 minutes. Refer to your Section 1A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term from Section 1A.

1.	Power BI Desktop saves analysis projects as .pbix files.
2.	The in-memory columnar database engine powering Power BI is called xVelocity .
3.	Power Query uses a language called M (M Language) to clean and transform data.
4.	Columnar databases store data grouped by column rather than by row, reducing unnecessary data reads.
5.	DAX stands for Data Analysis eXpression Language.
6.	The xVelocity engine is also known by its Excel-facing name, Power Pivot and VertiPaq , and its internal codename, .

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each Power BI component.

Component	
A. Power BI Desktop	2.
B. Power BI Service	4.
C. Power Query Editor	3.
D. Power BI Mobile	6.

Description
1. In-memory engine that hosts the data model (also called VertiPaq)
2. Free Windows app for building semantic models and authoring reports
3. Provides M-language tooling to ingest, clean, and reshape data
4. Cloud-based SaaS version of Power BI

E. xVelocity Engine 1.	5. Expression language used to add formulas and calculations to the model
F. DAX 5.	6. Lets users view reports on iOS, Android, or Windows mobile devices

Part 3: Short Answer (~10 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

A classmate says: "Power BI is just a chart-making tool, like Excel graphs." How would you correct this? What does Power BI actually do end-to-end?

Power BI is much more than a chart-making tool. It handles the entire data analysis workflow end-to-end — ingesting data from dozens of sources, cleaning and transforming it through Power Query, building a structured semantic model with defined relationships, and then creating interactive visualizations on top of that model. Excel graphs are static visuals tied to spreadsheet data; Power BI reports are dynamic, filterable, and built on a fully structured data model.

Question 2

Why does Power BI use a columnar database engine (xVelocity) instead of a traditional row-based database? What advantage does this give analysts working with large datasets?

Traditional row-based databases read entire rows off disk even when you only need a few columns, wasting time and memory. xVelocity stores data grouped by column, so when an analyst runs a query that only needs 3 out of 20 columns, only those 3 columns are read. This, combined with in-memory storage and compression, allows Power BI to aggregate millions of rows almost instantly — far faster than reading from disk row by row.

Question 3

Your manager asks you to share a Power BI report with a colleague who works remotely on a tablet. Which Power BI ecosystem component(s) are involved, and what does each one do?

Two components are involved. First, Power BI Service — the cloud-based version where the finished report gets published so it's accessible online. Second, Power BI Mobile — the app the colleague would use on her tablet to view the report in a mobile-friendly format. Power BI Desktop is where the report was built, but sharing and remote access happen through the Service and Mobile app.

Question 4

DAX adds four types of calculations to a Power BI model. Name all four and describe each in one sentence.

Calculation Type	What it does (your description)
Calculated Columns	Adds a new column to a table by computing a value row
Measures	Dynamic calculations that are evaluated on demand within a filter
Calculated Tables	Creates a new table in the model using a DAX expression

Row Filters	Controls which rows of data a specific user or role is allowed to see

Part 4: Scenario Analysis (~7 minutes)

For each scenario, identify the most appropriate Power BI component or concept from Section 1A. Write the component name and a brief explanation.

Scenario	Your Answer (Component + Why)
1. A sales analyst needs to load a messy CSV file, remove duplicates, and rename columns before building a report.	Power Query Editor — this is exactly what Power Query is designed for: connecting to data sources like CSVs and applying step-by-step transformations like removing duplicates and renaming columns before the data enters the model
2. A team wants to calculate each product's year-to-date revenue total -- a value that changes based on which filters are applied on the report page.	DAX Measure — measures are evaluated dynamically within whatever filter context exists on the report page, making them perfect for calculations that need to respond to slicers and filters
3. An executive needs to view the company dashboard while travelling. She only has her iPhone.	Power BI Mobile — the mobile app connects to the Power BI Service and displays reports in a mobile-friendly format on iOS, Android, and Windows devices
4. A developer asks why Power BI can aggregate 50 million rows of sales data in under a second when the server only has 16 GB of RAM.	xVelocity Engine (columnar storage + compression) — columnar storage only reads the columns needed, and heavy compression allows massive datasets to fit entirely in memory, making aggregations extremely fast without needing to read from disk
5. Your team has finished the semantic model and wants non-technical stakeholders to access and share reports through a web browser.	Power BI Service — the cloud-based SaaS platform where finished reports are published and shared, allowing anyone with a browser to access and interact with reports without needing Power BI Desktop installed

Bonus / Reflection (if time allows)

Think back to earlier in this course when you used Excel, SQL, and Python for data analysis. In 3-5 sentences, explain how Power BI fits into that workflow. What does it add that those tools cannot provide on their own?